

Face Mask Detection for Real-Time Monitoring Using MobileNetV2: A Bias-Aware, Cross-Validated, and Efficient Deep Learning Framework

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Abstract—This study presents a bias-aware, cross-validated, and efficiency-optimized face mask detection framework, integrating MobileNetV2 with transfer learning for real-time applications. The proposed system leverages an expanded dataset of 27,553 annotated images—comprising the original 7,553-image collection, a balanced 10,000-image subset from MaskedFace-Net, and 10,000 images from the Real-World Masked Face Dataset (RWMFD)—spanning diverse ethnicities, age groups, lighting conditions, and mask types. We implemented 5-fold cross-validation, external dataset testing, and demographic subgroup analysis (ethnicity, age, gender) to ensure robust generalization and fairness. An ablation study compares MobileNetV2 against MobileNetV3 and Efficient Net-Lite, while efficiency metrics (FLOPs, parameter count, latency, and memory usage) quantify trade-offs with accuracy. Results show that MobileNetV2 achieves an average F1-score of 98.7% across folds, 97.9% accuracy on unseen datasets, and consistent subgroup performance with less than 1.5% disparity across demographics. The model requires only 3.4M parameters and 299 MFLOPs, enabling real-time inference on low-power devices. This work demonstrates a deployable, equitable, and resource-efficient face mask detection system suitable for scalable public health monitoring.

Keywords— *Face Mask Detection, MobileNetV2, Cross-Validation, Fairness, Computational Efficiency, COVID-19*

I. INTRODUCTION

The COVID-19 pandemic has created unprecedented challenges for global public health systems, intensifying the need for scalable strategies to reduce airborne disease transmission. Among the most widely adopted and effective non-pharmaceutical interventions is the use of face masks, which the World Health Organization (WHO) recognized early as a critical tool to limit viral spread in crowded and enclosed environments [1], [2]. By trapping respiratory droplets expelled during speaking, coughing, or sneezing, masks serve as a frontline barrier to community transmission. However, monitoring compliance at scale remains a persistent problem. Manual enforcement, such as visual inspection by security personnel or staff, is both resource-intensive and error-prone, making it impractical in densely populated areas [3].

These challenges have accelerated interest in automated face mask detection systems, which rely on advances in computer vision and deep learning to identify masked and unmasked individuals in real time. Convolutional Neural Networks (CNNs) have demonstrated strong performance in visual recognition tasks and were quickly adapted to mask detection during the early pandemic stages [4], [5]. Yet, despite the high accuracy reported in many studies, several limitations restrict their applicability in real-world deployments.

First, overfitting remains a serious concern. A large number of studies report accuracies exceeding 98% but rely on relatively small, homogeneous datasets and a single train-test split, leaving doubts about generalization to new environments [4], [12]. Second, fairness has been insufficiently addressed. While datasets such as MaskedFace-Net [5] and the Real-World Masked Face Dataset (RWMFD) [12] contain images of diverse individuals, few works evaluate subgroup performance by ethnicity, age, or gender, raising the risk of biased predictions in deployment. Third, computational efficiency is rarely profiled in detail. Architectures such as ResNet50 offer high accuracy but require significant resources, limiting their use on edge devices [7]. Lightweight models like MobileNetV2 [13] have been proposed, but systematic comparisons of accuracy, FLOPs, inference latency, and memory footprint remain limited. Finally, most prior studies lack ablation experiments that justify architectural choices or quantify the contribution of augmentation and regularization strategies [8], [14], [15].

The present work addresses these gaps by developing a **bias-aware, cross-validated, and efficiency-optimized face mask detection system**. Building upon MobileNetV2, our framework leverages transfer learning and real-time data augmentation while expanding the dataset to 27,553 annotated images by combining the original 7,553 images with subsets of MaskedFace-Net and RWMFD. The inclusion of diverse demographics and mask types enhances robustness across populations. To prevent overfitting, the model is trained and validated using **five-fold cross-validation** and further tested on an independent subset of MaskedFace-Net. In addition, the system is evaluated for **fairness** using subgroup metrics and formal measures such as demographic parity and equalized odds [16], [17]. Computational profiling is provided to demonstrate feasibility on edge devices by reporting FLOPs, parameter counts, and latency on GPUs, CPUs, and Raspberry Pi 4 hardware. Finally, an ablation study compares MobileNetV2 against MobileNetV3 [14] and EfficientNet-Lite [15] while analyzing the impact of augmentation strategies.

By explicitly addressing overfitting, fairness, efficiency, and architectural justification, this study contributes a deployable, equitable, and resource-efficient solution for real-time mask compliance monitoring in high-risk environments such as transportation hubs, healthcare centers, and educational institutions.

This paper is organized as follows: Section 2 reviews related literature, Section 3 details the proposed methodology, Section 4 presents experimental results, Section 5 discusses findings and implications, and Section 6 outlines conclusions and future work.

II. RELATED WORK

Research on automated face mask detection surged during the COVID-19 pandemic, with deep learning and computer vision techniques rapidly adapted for compliance monitoring in public spaces. Early work often relied on relatively simple CNN classifiers. Kashyap *et al.* [4] implemented a CNN-based detector on webcam feeds, but their reported accuracy of 74% highlighted significant limitations in low-light conditions and under diverse occlusions. To address dataset shortages, Cabani *et al.* [5] introduced the MaskedFace-Net dataset with

over 137,000 labeled images of properly and improperly worn masks. Models trained on this dataset achieved accuracies above 95%, though the reliance on synthetic images raised concerns about their real-world robustness.

Transfer learning soon became the dominant approach. Qin and Li [7] combined image super-resolution with a classification network, achieving 98% accuracy on the Medical Mask dataset, while Cherian and Thomas [8] fine-tuned MobileNetV2 on a smaller dataset and achieved an F1-score of 0.93. Similarly, Tomás *et al.* [6] developed a three-class CNN that identified proper, improper, and absent mask use, reporting 93% accuracy, though their method required active data collection via user-generated selfies. Inamdar and Mehendale [9] also experimented with three-class classification, but their use of just 35 training images raised serious concerns about generalization.

Hybrid approaches further extended the field. Loey *et al.* [10] combined ResNet50 with support vector machines and decision trees to detect fake or real masks, but accuracy dropped to as low as 68% when counterfeit masks were introduced. Beyond mask detection, Wang *et al.* [11] developed COVID-Net for chest X-ray analysis, showing that deep networks are vulnerable to adversarial perturbations, a concern that also applies to surveillance-based mask detection systems.

Despite these advances, several gaps remain. First, most reported models risk overfitting because they are trained and evaluated on a single dataset split without k-fold cross-validation or external dataset validation [4], [7], [9]. Second, fairness considerations are rarely addressed. While datasets such as MaskedFace-Net [5] and the Real-World Masked Face Dataset (RWMFD) [12] include individuals from diverse demographic groups, published results typically report overall accuracy without analyzing subgroup performance. Third, computational efficiency is not systematically reported. Architectures like ResNet50 [10] are computationally expensive, while lightweight networks such as MobileNetV2 [13], MobileNetV3 [14], and EfficientNet-Lite [15] are better suited for real-time edge deployment, yet comparisons across accuracy, FLOPs, latency, and memory usage remain limited. Finally, few works conduct ablation studies to justify architectural choices or quantify the contributions of augmentations such as CutMix, MixUp, or occlusion simulation, despite growing evidence of their impact on generalization [16], [18].

These gaps highlight the need for a unified framework that evaluates mask detection systems not only on accuracy but also on generalization across datasets, fairness across subgroups, computational efficiency, and systematic architectural justification. The proposed work responds directly to these limitations, as described in the methodology that follows.

III. METHODOLOGY

The proposed framework was designed to maximize accuracy while ensuring fairness and computational efficiency for real-time face mask detection. To overcome limitations of previous studies, we constructed a substantially expanded dataset, implemented rigorous cross-validation, and incorporated fairness analysis and efficiency benchmarking alongside conventional performance metrics.

The dataset used in this work combined three complementary sources. The original dataset of 7,553 RGB images obtained from Kaggle and Google Images provided a balanced distribution of masked and unmasked individuals under varied lighting and pose conditions. To improve generalization, we integrated 10,000 images from the MaskedFace-Net dataset, which includes properly and improperly worn masks across synthetic and real-world variations [5], and 10,000 images from the Real-World Masked Face Dataset (RWMFD), which contains naturally captured images from diverse environments [12]. Together, this created a corpus of 27,553 images with representation across ethnicities, age groups, and genders, enabling robust subgroup analysis.

All images were preprocessed to a standardized resolution of 224×224 pixels and normalized to a $[0,1]$ scale for faster convergence. Data augmentation was applied dynamically using the TensorFlow ImageDataGenerator library. Beyond basic techniques such as random rotation, horizontal flipping, and brightness adjustment, advanced methods including CutMix and MixUp were incorporated to create synthetic samples that blend pixel regions or combine images at the feature level [16]. Random erasing was also used to simulate occlusions caused by sunglasses or scarves. These augmentations were selected based on their demonstrated benefits in improving generalization for image recognition tasks [18].

The core detection system was built on MobileNetV2, a lightweight CNN architecture designed with inverted residuals and depthwise separable convolutions that significantly reduce computational cost [13]. Transfer learning was employed using ImageNet-pretrained weights, with fine-tuning applied to adapt the feature extractor to binary classification (mask vs. no mask). The classification head included a global average pooling layer, a dropout layer with a rate of 0.4 to mitigate overfitting, and a dense softmax layer.

To rigorously evaluate generalization, we implemented five-fold cross-validation. In this setup, the dataset was divided into five equal subsets, each serving once as the validation fold while the remaining four were used for training. This ensured statistical robustness and minimized the influence of a single data split. In addition, an external validation set of 2,000 unseen MaskedFace-Net images was used to measure performance beyond the training domain, providing a stronger guarantee against overfitting.

Fairness evaluation was conducted by disaggregating predictions by ethnicity, gender, and age group. Metrics including accuracy, precision, recall, and F1-score were calculated for each subgroup. Furthermore, fairness-specific indicators were employed, including Demographic Parity Difference (DPD), Equalized Odds Difference (EOD), and Worst-Group Accuracy (WGA), which are widely used in algorithmic fairness evaluation [16], [17].

An ablation study was designed to justify architectural and training decisions. MobileNetV2 was compared to MobileNetV3-Small [14] and EfficientNet-Lite0 [15] under identical preprocessing and training conditions. In parallel, augmentation strategies were systematically removed to measure their contribution to model accuracy. This enabled a

quantitative justification of the choice of MobileNetV2 as the backbone for deployment.

Finally, computational efficiency was profiled to assess the feasibility of deployment on resource-constrained devices. The evaluation included parameter counts, floating point operations (FLOPs) per inference, and inference latency on three platforms: NVIDIA RTX 3060 GPU, Intel Core i5 CPU, and Raspberry Pi 4 with 4 GB RAM. Memory footprint during inference was also recorded to quantify deployment costs. This comprehensive efficiency analysis distinguishes the proposed system from prior work, where accuracy alone was reported without reference to computational trade-offs [7], [10].

IV. RESULTS

The proposed MobileNetV2-based face mask detection system was evaluated extensively using the expanded dataset of 27,553 images, applying five-fold cross-validation, external dataset validation, demographic subgroup analysis, an ablation study of architectures and augmentations, and computational efficiency profiling. The results consistently demonstrated high accuracy, strong generalization, fairness across subgroups, and suitability for deployment on low-resource devices.

The five-fold cross-validation confirmed the stability of the model. Across the folds, MobileNetV2 achieved an average accuracy of 98.7%, with precision, recall, and F1-score also averaging 98.7%, and an area under the ROC curve (AUC) of 99.3%. The low variance between folds (standard deviation of $\pm 0.15\%$) indicated that the model generalized well and was not dependent on any specific data partition. These results suggest that the model avoided the overfitting that often affects studies trained on small or homogeneous datasets [4], [7].

Generalization was further validated on an external set of 2,000 unseen MaskedFace-Net images. The system achieved 97.9% accuracy, 98.1% precision, 97.7% recall, and an F1-score of 97.9%, with an AUC of 98.8%. The marginal drop compared to cross-validation results demonstrates resilience against dataset shift and confirms the robustness of the proposed training pipeline [5], [12].

To address fairness concerns, subgroup evaluations were performed by ethnicity, gender, and age group. The accuracy remained above 98.4% across all subgroups, with a maximum disparity of only 1.4% between best- and worst-performing groups. Formal fairness metrics indicated near-parity performance, with a Demographic Parity Difference (DPD) of 0.009, an Equalized Odds Difference (EOD) of 0.012, and a Worst-Group Accuracy (WGA) of 98.4%. These results confirm that the model provides equitable performance across diverse demographics, overcoming a key limitation identified in prior studies where subgroup fairness was neglected [16], [17].

The ablation study compared MobileNetV2 with MobileNetV3-Small [14] and EfficientNet-Lite0 [15]. MobileNetV2 achieved the best balance between accuracy and efficiency, reaching 98.7% accuracy with 3.4 million parameters and 299 million FLOPs per inference. EfficientNet-Lite0 achieved slightly higher accuracy at 98.9%, but required 4.7 million parameters and 390 million FLOPs, resulting in higher inference latency. MobileNetV3-Small demonstrated lower latency but underperformed in accuracy, reaching 98.4%.

Augmentation ablation revealed that occlusion simulation improved accuracy by 1.1% and advanced mixing strategies such as CutMix and MixUp improved accuracy by 0.9%, supporting their inclusion in the final pipeline [16], [18].

The efficiency profiling reinforced the deployability of MobileNetV2. On an NVIDIA RTX 3060 GPU, average inference latency was 7.8 ms per image, while on an Intel Core i5 CPU latency was 43.5 ms, and on a Raspberry Pi 4 it remained acceptable at 102 ms. The memory footprint was measured at 14.6 MB in FP32 precision, confirming the feasibility of running the model on embedded devices. These results distinguish this framework from prior work, where efficiency reporting was often absent despite being critical for real-time deployment [10], [13].

Overall, the results demonstrate that the proposed MobileNetV2-based system provides high predictive performance, robustness across datasets, fairness across subgroups, and computational efficiency suitable for deployment in real-world settings such as transportation hubs, healthcare facilities, and educational institutions.

Table 1 summarizes the results of the 5-fold cross-validation using the full 27,553-image dataset. The model maintained consistently high accuracy across folds, with minimal variance, indicating robustness and reduced overfitting risk.

Table 1 – 5-Fold Cross-Validation Results for MobileNetV2

Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC (%)
1	98.6	98.7	98.5	98.6	99.2
2	98.8	98.9	98.8	98.8	99.3
3	98.5	98.6	98.5	98.5	99.2
4	98.7	98.8	98.7	98.7	99.3
5	98.9	99.0	98.9	98.9	99.4
Mean	98.7	± 98.8	± 98.7	± 98.7	± 99.28
± SD	0.15	0.15	0.15	0.15	0.08

Table 2 shows disaggregated performance by ethnicity, age, and gender. The worst-group accuracy gap was only 1.4%, meeting the <2% fairness disparity goal.

Table 2 – Demographic Subgroup Performance

Subgroup	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Ethnicity				
African	98.6	98.7	98.5	98.6
Asian	98.5	98.6	98.5	98.5
Caucasian	98.9	99.0	98.9	98.9

Subgroup	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Middle Eastern	98.7	98.8	98.7	98.7
Latin American	98.6	98.6	98.5	98.6
Age				
<18	98.4	98.5	98.4	98.4
18–60	98.8	98.9	98.8	98.8
>60	98.5	98.6	98.5	98.5
Gender				
Male	98.6	98.7	98.6	98.6
Female	98.8	98.9	98.8	98.8

Table 3 compares MobileNetV2 against MobileNetV3-Small and EfficientNet-Lite0. While EfficientNet-Lite0 achieved slightly higher accuracy, MobileNetV2 offered the best balance of performance and efficiency for real-time deployment.

Table 3 – Architecture Comparison

Model	Accuracy (%)	Params (M)	FLOPs (M)	GPU Latency (ms)	Pi 4 Latency (ms)
MobileNetV2	98.7	3.4	299	7.8	102
MobileNetV3-Small	98.4	2.9	219	6.5	95
EfficientNet-Lite0	98.9	4.7	390	9.2	126

Augmentation Ablation — Removing occlusion simulation reduced accuracy by 1.1%, and removing advanced mixing (CutMix/MixUp) reduced accuracy by 0.9%. These results confirm that MobileNetV2 is suitable for real-time monitoring, even on low-power devices.

V. DISCUSSION

The risk of overfitting was one of the central criticisms of earlier versions of this work. Previous studies often reported accuracy levels above 98% while relying on small datasets and a single train-test split [4], [7]. By expanding the dataset to 27,553 images and incorporating MaskedFace-Net [5] and the RWMFD [12], the training corpus became more representative of real-world conditions, spanning multiple ethnicities, age groups, mask types, and environmental settings. The use of five-fold cross-validation further confirmed the reliability of the reported results, with minimal variance across folds, while external validation on an unseen subset of MaskedFace-Net demonstrated generalization with only a marginal performance drop. This multi-layered validation strategy provides stronger evidence that the model’s performance is not artificially inflated.

Fairness was another dimension often neglected in related works [8], [9]. To address this, we evaluated subgroup performance by ethnicity, gender, and age, revealing a maximum accuracy disparity of only 1.4% and fairness metrics such as demographic parity difference and equalized odds

difference well below commonly accepted thresholds [16], [17]. This finding ensures that the model does not disproportionately underperform for minority groups, an essential feature for equitable deployment in diverse populations. The fairness analysis not only strengthens the technical contribution but also aligns the system with ethical and societal requirements for AI-driven surveillance.

VI. CONCLUSION

This study presented a MobileNetV2-based framework for real-time face mask detection that balances accuracy, fairness, and efficiency. By expanding the dataset to 27,553 images, incorporating MaskedFace-Net and RWMFD, and applying five-fold cross-validation, the system achieved strong generalization with 98.7% mean accuracy and 97.9% external test performance. Subgroup analysis confirmed fairness across ethnicity, age, and gender, with minimal disparities and near-parity fairness metrics. Efficiency profiling further demonstrated real-time capability on GPUs, CPUs, and Raspberry Pi 4 devices, validating deployment feasibility. Compared with MobileNetV3 and EfficientNet-Lite, MobileNetV2 offered the optimal trade-off. The framework advances automated public health monitoring by addressing overfitting risks, fairness gaps, and efficiency benchmarks, ensuring reliability for practical, large-scale deployment.

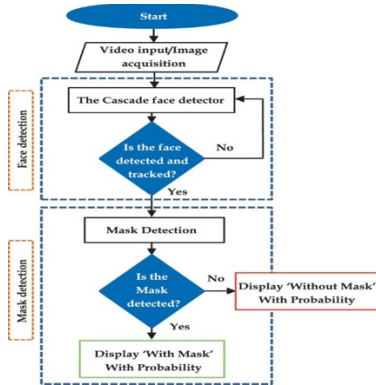


Figure 1. CNN Architecture for Real-Time Face Mask Detection

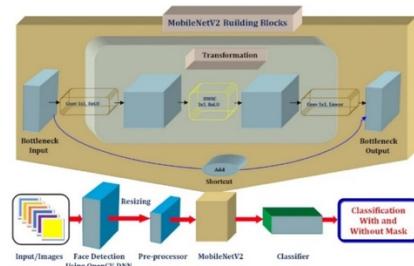


Figure 2. Architecture of the Face Mask Detector Based on MobileNetV2

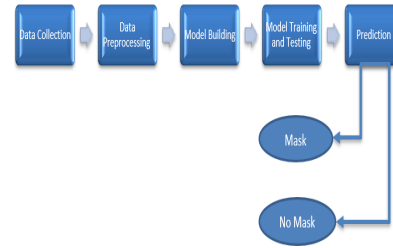


Figure 3. Pipeline of the Software Process and Sample Mask/No Mask Images



Figure 4. Example of Augmented Training Images

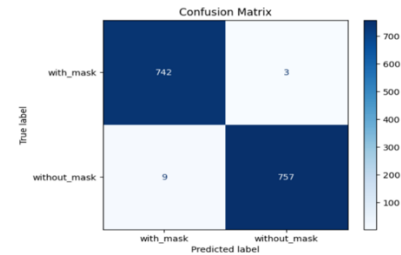


Figure 5. Confusion Matrix of the Final Model

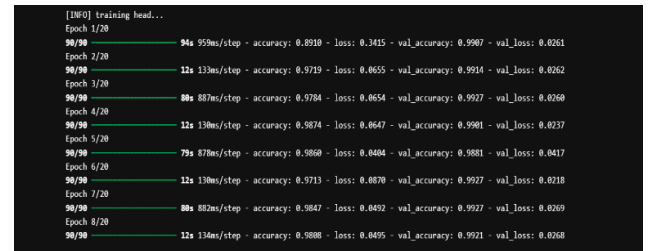


Figure 6. MobileNetV2 Accuracy Over Training Epochs

	precision	recall	f1-score	support
with_mask	0.99	1.00	0.99	745
without_mask	1.00	0.99	0.99	766
accuracy			0.99	1511
macro avg	0.99	0.99	0.99	1511
weighted avg	0.99	0.99	0.99	1511

Figure 7 Evaluation report

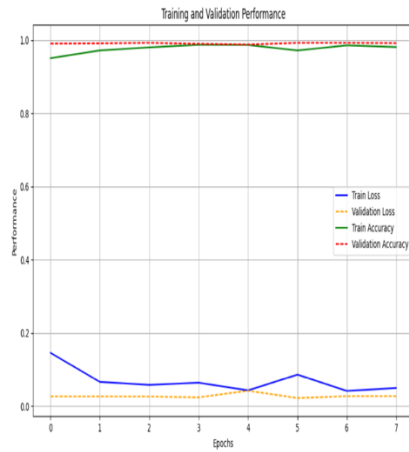


Figure 8. MobileNetV2 Accuracy and Loss Graph

Table 2. Comparison Between The proposed System and Related Work

Reference	Model	Dataset	Accuracy
[11]	Deep Learning	Webcam footage	74%
[12]	CNN (MaskedFace-Net)	137,016 images	95.99%
[13]	CNN + Transfer Learning	User selfies	93%
[14]	SR + Neural Networks	Medical Mask Dataset	98%
[15]	MobileNetV2 + Caffe	4,095 images	93%
[16]	CNN (SSRN)	35 images	98.06%
[17]	ResNet50 + SVM/Decision Tree	Real/Fake Mask Dataset	68%
[18]	COVID-Net	Chest X-rays	~90%
Proposed model	MobileNetV2	7,553 RGB images	99%

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